

MASENO 2025 KCSE PREDICTION



CONFIDENTIAL EXAM

232/1

PHYSICS

PAPER 1 (THEORY)

TIME: 2 HOURS



NAME.....

SCHOOL..... SIGN.....

INDEX NO..... ADM NO.....

Instructions to Candidates

- ❖ Write your name and index number in the spaces provided above.
- ❖ Sign and write the date of the examination paper. This paper consists of two sections; A and B
- ❖ Answer **ALL** the questions in the spaces provided in the question paper.
- ❖ **ALL** working **MUST** be clearly shown where necessary.
- ❖ Mathematical tables and silent electronic calculators may be used. Use $g=10\text{N/Kg}$

FOR EXAMINER'S USE ONLY

SECTION	QUESTIONS	MAXIMUM SCORE	CANDIDATE'S SCORE
A	01-09	25	
B	10	11	
	11	11	
	12	06	
	13	07	
	14	11	
	15	09	
TOTAL SCORE		80	

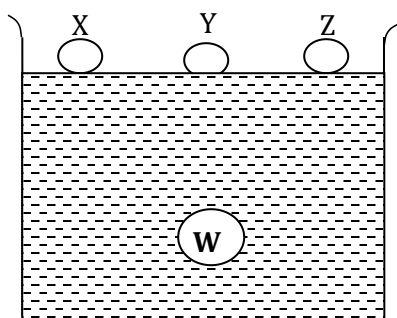
SECTION A: 25 MARKS

1. (a) The vernier calipers shows the reading of 5.38cm. Draw the diagram showing this.

(2 marks)

- (b) If the vernier calipers used had an error of 0.05cm, find the actual reading **(1 mark)**

2. Use the surface molecules X, Y, Z and inner molecule W shown below to explain why the surface of a liquid is under tension **(3 marks)**



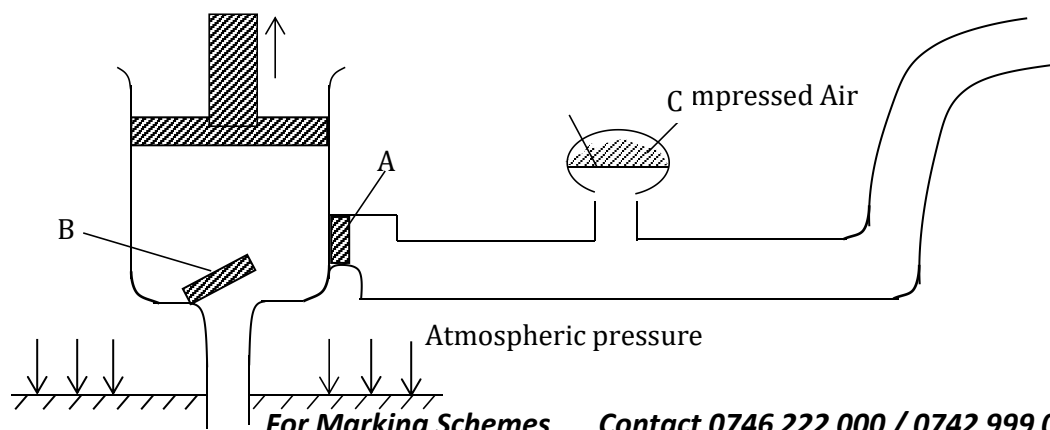
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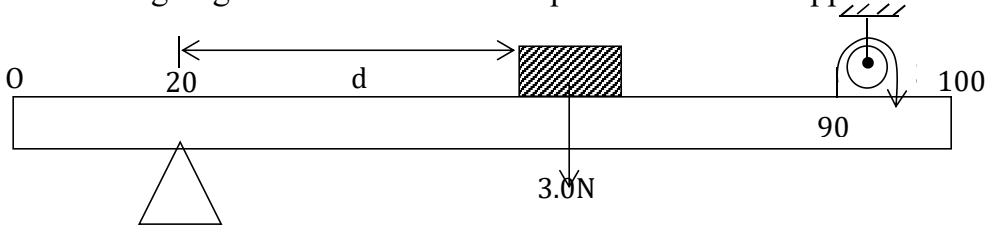
3. The diagram below shows a force pump used to draw water from a well.



Explain how the continuous flow of water through outlet is achieved (2marks)

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4. A uniform meter rule weighing 1.5 N is in a state of equilibrium when supported as shown below.



Calculate the distance, d (3 marks)

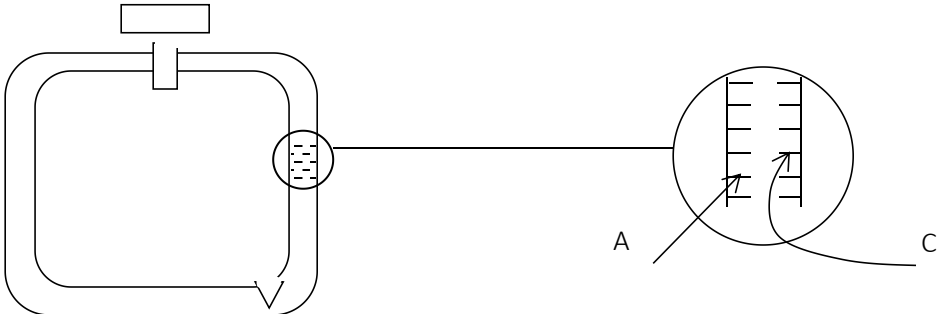
5. Oil flows through a 14cm internal diameter pipe at an average velocity of 10m/s. Find the flow rate in SI unit (3 marks)

6. (a) State the Newton’s second law of motion (1 mark)

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(b)A 90kg water skier is being pulled by a speed boat. The force causes her to accelerate at 4m/s². Calculate the accelerating force. **(2 marks)**

7. The diagram below shows a simple diagram of a vacuum flask with an enlarged view of the part in a circle.



(a)What is the material of item labelled A and C? **(1 mark)**

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(b)Explain why A and C is effective in reducing heat transfer. **(2 marks)**

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8. **(a)** State Hooke’s law **(1 mark)**

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(b) A mass of 200g is suspended from the lower end of a spring. If the reading when no mass is hanging on the spring is 45.5 cm and the reading after the mass is suspended is 49.5cm. Calculate the spring constant. **(3 marks)**

9. State one assumptions made during determination of the size of oil molecule in the oil drop experiment.
(1 marks)

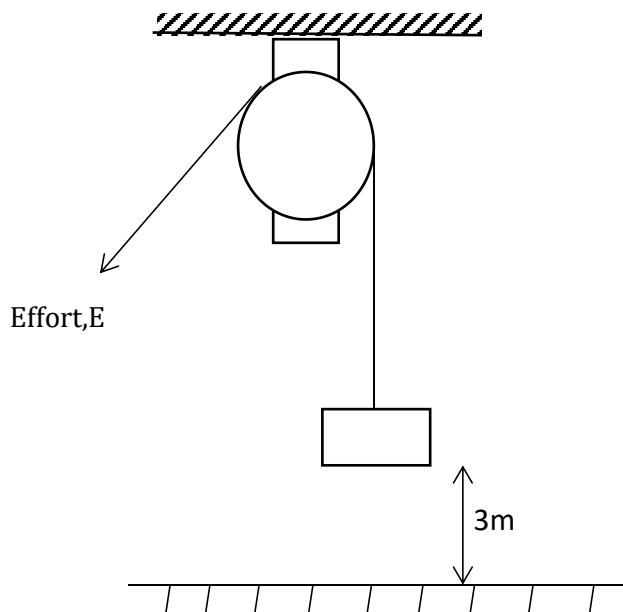
SECTION B: 55 MARKS

10. (a) Define work and state its SI unit (2 marks)

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- (b) A force of 15N is applied on a body causing it to move through a distance of 2m on a surface that offers a friction of 5N Determine the useful work done (3 marks)

- (c) A man uses a pulley shown in the diagram below to lift a load of 100N through a distance of 3m.



(i) If the friction is negligible, what is the value of effort E (2 marks)

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(ii) Determine the work done in raising the bucket through a height of 3m (2 marks)

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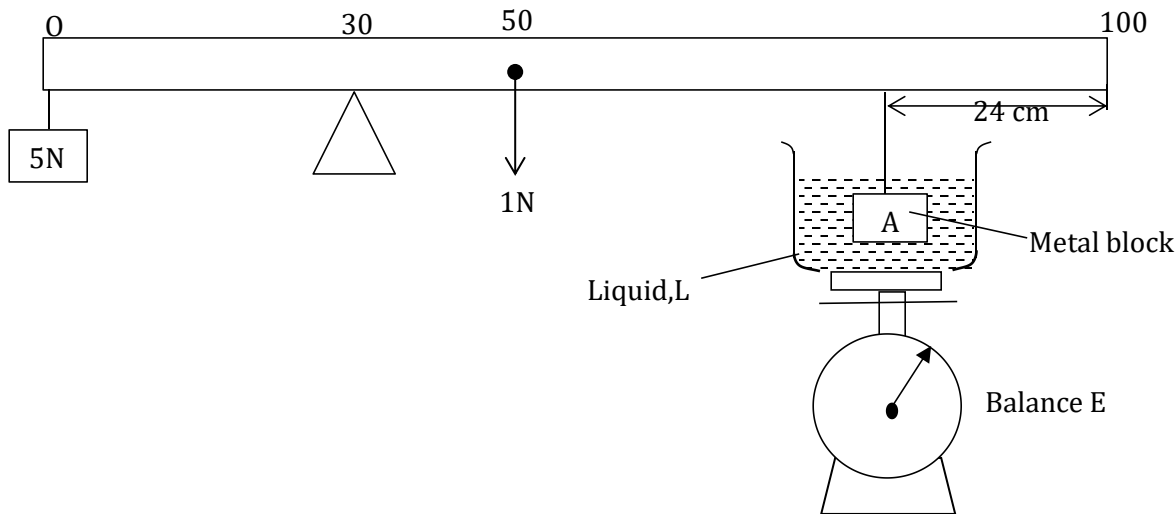
(iii) Determine the work done by the effort,E (2 marks)

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11.The figure below shows a lever arrangement with a uniform meter rule of weight 1N balanced by a knife edge at the 30 cm mark. The 5N weight at 0cm mark balances metal block A of volume $2.8 \times 10^{-3} \text{ m}^3$ which has been immersed in liquid L. The weight of the beaker is 2.0N and that of liquid is 3.0N. Balance E reads 15N.



Determine:

(a) The apparent weight of the metal block A (2 marks)

(b) Upthrust on the metal block due to the liquid L (2 marks)

(c) The density of liquid L (2 marks)

(d) The weight of the metal block in air (2 marks)

(e) The density of the solid (2 marks)

(f) Why does hydrometer have a narrow stem (1 mark)

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12. A body is projected horizontally at an initial velocity of 120 m/s from a cliff 300m tall. Calculate;

(a) Time of flight (2 marks)

(b) The range (2 marks)

(c) Vertical velocity at which the body strikes the ground (2 marks)

13. A stone is whirled with uniform speed in a horizontal circle having a radius of 16cm. it takes the stone 12 seconds to describe an arc of length 8cm.

Calculate;

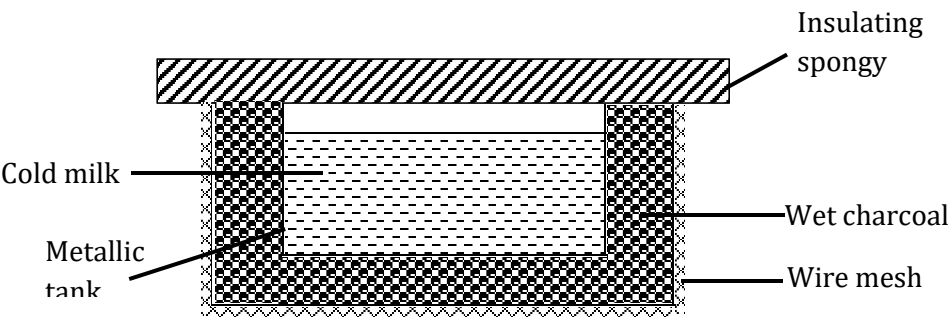
(a) The angular displacement (2 marks)

(b) The angular velocity
(1 mark)

(c) The linear velocity of the stone (2 marks)

(d) The periodic time (2 marks)

14.(a)The diagram below shows a charcoal refrigerator used to cool milk to low temperature.



Explain how the cooling takes place in this refrigerator (2 marks)

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(b) Give a reason why charcoal is suitable for cooling process (1 mark)

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(c)(i) Define the term heat capacity and state its SI unit (2 marks)

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(ii) An electric heater rated 15A, 240V is used to heat 2kg of water from 20°C to 100°C in 160 seconds.
Determine the specific heat capacity of water. (2 marks)

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(iii) Explain why the value calculated in C (ii) an estimate **(2 marks)**

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(d) Explain how boiling point of a substance is affected by altitude **(2 marks)**

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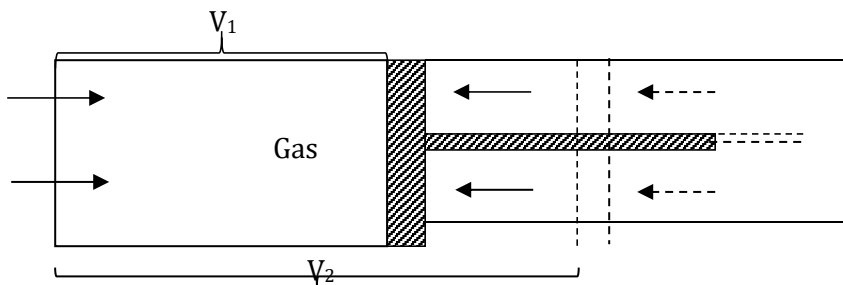
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15.(a) Define absolute zero. **(1 mark)**

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(b) A gas is enclosed inside a cylinder which is fitted with a frictionless piston. Initially, the gas has a volume V_1 and is in equilibrium with an external pressure P . The piston is then pulled backward until the volume becomes V_2 which is twice V_1 .



(i) State the change in pressure. **(1 mark)**

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(ii) Explain the change in pressure in (i) (2 marks)

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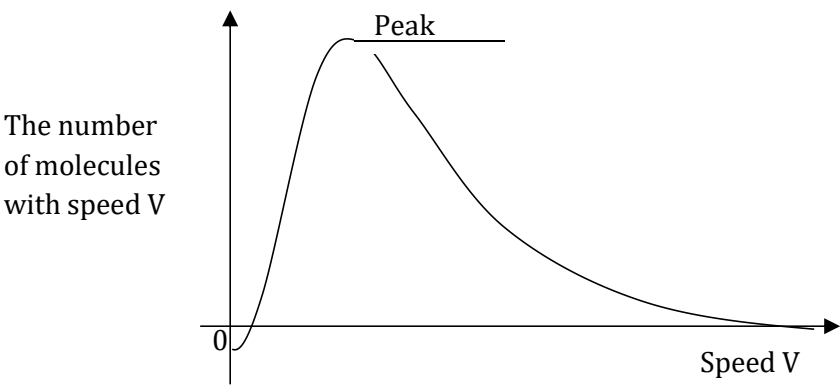
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(iii) State the law that governs the above experiment. (1 mark)

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(c) The graph below shows the distribution of speeds of a gas at a particular temperature



State how the peak of the graph would change if temperature is increased. (1 mark)

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(d) The volume of a fixed mass of a gas reduced from 500cm³ to 300cm³ at constant pressure. The initial temperature was 90k. Determine the final temperature. (3marks)